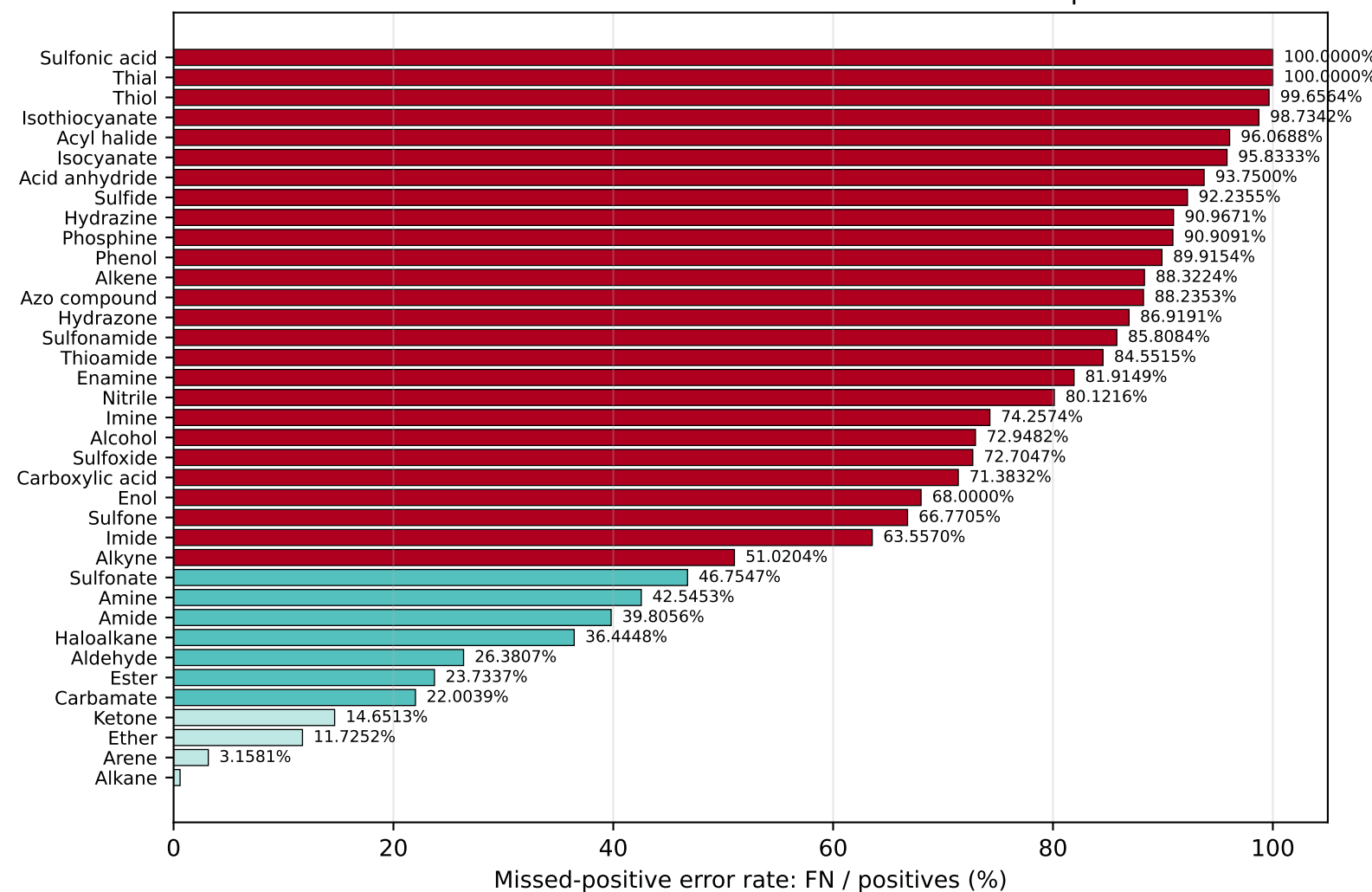
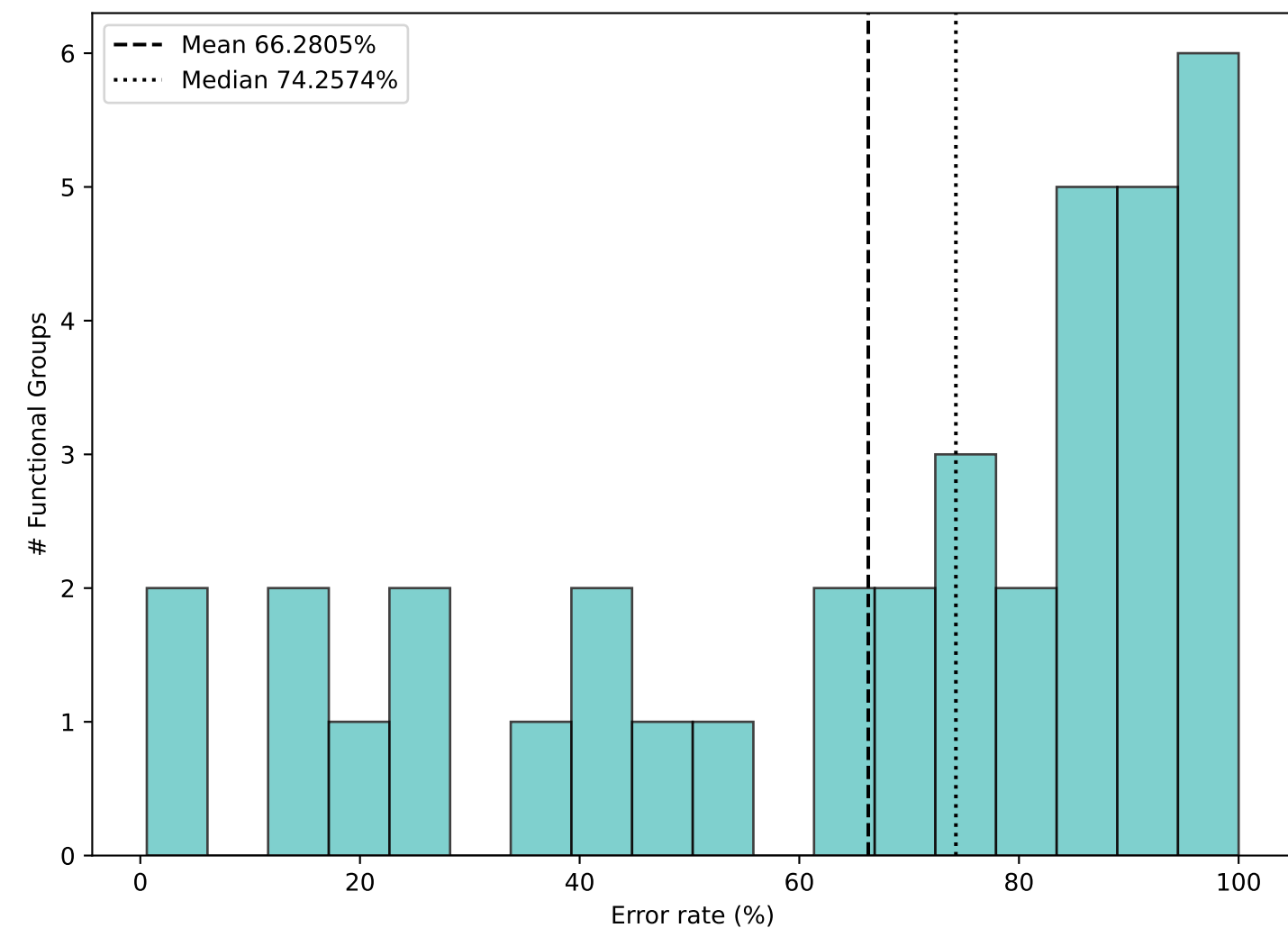


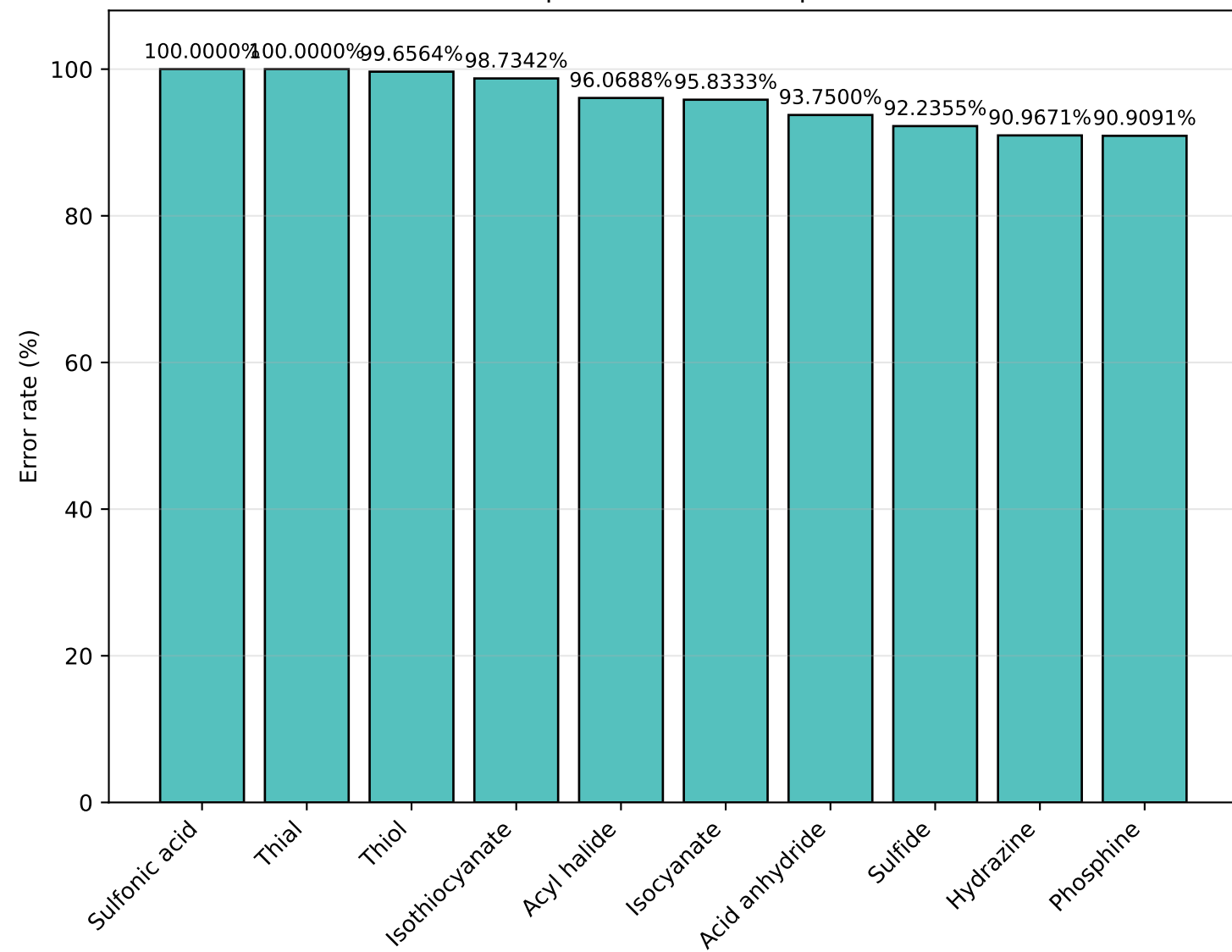
XGBoost C-NMR Test Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



XGB00ST C-NMR TEST SET ERROR ANALYSIS

Global metrics:

F1 micro: 0.8020
F1 macro: 0.4450
Precision micro: 0.8851
Recall micro: 0.7327
Hamming accuracy: 0.9531

Error-rate statistics:

Mean: 66.2805%
Median: 74.2574%
Std: 30.8259%
Max: 100.0000% (Sulfonic acid)
Min: 0.5909% (Alkane)

Problem classes:

Severe (>50%): 26 groups
High (20-50%): 7 groups
Moderate (5-20%): 2 groups
Low (<=5%): 2 groups